

Multiple Choice (10 marks)

1. Which of the following does authentication aim to accomplish?
 - (a) Restrict what operations/data the user can access
 - (b) Determine if the user is an attacker
 - (c) Flag the user if he/she misbehaves
 - (d) Determine who the user is
2. _____ is a popular tool used for network analysis in multiprotocol diverse network.
 - (a) Snort
 - (b) SuperScan
 - (c) Burp Suit
 - (d) EtterPeak
3. _____ is a process of wireless traffic analysis that may be helpful for forensic investigations or during troubleshooting any wireless issue.
 - (a) Wireless Traffic Sniffing
 - (b) WiFi Traffic Sniffing
 - (c) Wireless Traffic Checking
 - (d) Wireless Transmission Sniffing
4. _____ is the central node of 802.11 wireless operations.
 - (a) WPA
 - (b) Access Point
 - (c) WAP
 - (d) Access Port
5. _____ is a widespread app's coding mistake made by developers which could be exploited by an attacker for gaining access or malfunctioning your system.
 - (a) Memory leakage
 - (b) Buffer-overflow
 - (c) Less processing power
 - (d) Inefficient programming

True / False (10 marks)

Answer True or False

6. True or False: Nessus is primarily designed for network intrusion detection and real-time traffic analysis.
7. True or False: IP address spoofing can be effectively detected by simply implementing a firewall on the network.
8. True or False: A digital signature relies on a public-key system for verifying the authenticity and integrity of digital messages.
9. True or False: Authorization aims to restrict what operations and data a user can access within a system.
10. True or False: WPA 3 is the most secure wireless security protocol currently available, offering advanced encryption and protection against brute-force attacks.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the implications of weak or non-existent authentication mechanisms in the context of presentation layer vulnerabilities in cybersecurity.
12. Discuss how blinding techniques mitigate timing attacks in cryptographic systems, emphasizing the relationship between ciphertext randomization and adversarial observations.

End of Paper